

Chapman–Enskog Theory

Pop Dan Ionut

September 15, 2026

Contents

1	A top-view of the Chapman-Enskog approach	2
1.1	Basic notation for relativistic hydrodynamics	2
1.2	Approximate solution to the Boltzmann equation	3
1.3	Dynamics of the macroscopic fields	5
2	Warm-up: effective collision terms	6
2.1	The non-relativistic case	6
2.2	The relativistic case	9
A	Supplementary Calculations	11
A.1	Shear-stress calculation	11

1 A top-view of the Chapman-Enskog approach

The Chapman–Enskog method provides a systematic way to derive reduced equations from kinetic models exhibiting a separation of spatio-temporal scales between microscopic and macroscopic dynamics.

This report seeks to simply conceptualize the method, explaining it in small steps, starting from a top-level picture, and working out a simplified example of the method.

1.1 Basic notation for relativistic hydrodynamics

We use the mostly-negative metric convention for the spacetime metric:

$$g_{\mu\nu} = \text{diag}(1, -1, -1, -1). \quad (1)$$

In relativistic hydrodynamics, we seek to describe the dynamics of, for example, a violent neutron star collision, or of the quark-gluon plasma created in heavy-ion collisions, in terms of the four-velocity $u^\mu(x)$, the local temperature $T(x)$, and the chemical potential $\mu(x)$. The four-velocity satisfies the normalization condition

$$u^2 := u^\mu u_\mu = 1. \quad (2)$$

There is a natural decomposition of any vector, or more generally any tensor, into components parallel and perpendicular to the four-velocity u^μ . For a vector A^μ , this decomposition takes the form

$$A^\mu = (u \cdot A)u^\mu + \Delta^\mu_\nu A^\nu, \quad (3)$$

where $\Delta^\mu_\nu = g^\mu_\nu - u^\mu u_\nu$ is the projection operator onto the subspace orthogonal to u^μ .

The spatial part of A^μ is denoted also by:

$$A^{\langle\mu\rangle} := \Delta^\mu_\nu A^\nu \quad (4)$$

A useful operation is to extract the spatial, symmetric and traceless part of a rank-2 tensor $A^{\mu\nu}$. This is denoted by:

$$A^{\langle\mu\nu\rangle} := \Delta^{\mu\nu}_{\alpha\beta} A^{\alpha\beta}, \quad (5)$$

where

$$\Delta^{\mu\nu}_{\alpha\beta} := \frac{1}{2} \left(\Delta^\mu_\alpha \Delta^\nu_\beta + \Delta^\mu_\beta \Delta^\nu_\alpha \right) - \frac{1}{3} \Delta^{\mu\nu} \Delta_{\alpha\beta} \quad (6)$$

is the projection operator onto the spatial, symmetric and traceless subspace orthogonal to u^μ .

Finally, in any relativistic field theory, we often need to integrate over the momentum space of particles. For a function $f(k^\mu)$, the Lorentz-invariant integration measure is given by:

$$\int dK f(k^\mu) = \int \frac{d^4k}{(2\pi)^4} 2\pi \delta(k^2 - m^2) \theta(k^0) f(k^\mu) \quad (7)$$

$$= \int \frac{d^3\mathbf{k}}{(2\pi)^3 2E_{\mathbf{k}}} f(k^\mu) \quad (8)$$

1.2 Approximate solution to the Boltzmann equation

The primary object of study in relativistic kinetic theory is the distribution function $f(x^\mu, k^\mu)$, which describes the density of particles at spacetime position x^μ with four-momentum k^μ . This already is a coarse-grained description of an underlying microscopic dynamics. Both quantum and classical microscopic theories can give rise to such a kinetic description.

The dynamics of the distribution function is governed by the relativistic Boltzmann equation, which takes the form

$$k^\mu \partial_\mu f(x^\nu, k^\nu) = C[f](x^\nu, k^\nu), \quad (9)$$

where $C[f]$ is the collision operator encoding the interactions between particles. We will not delve into the specific form of $C[f]$ here, only state that it is non-linear in the distribution function f . A collision term encoding binary interactions will be in fact quadratic in f .

We note that for particles of mass m , the distribution function f will be a proper mathematical distribution, being supported on the mass shell:

$$k^2 := k^\mu k_\mu = m^2. \quad (10)$$

The goal of obtaining hydrodynamic equations can be stated as follows: instead of describing a system in terms of the full distribution function $f(x^\mu, k^\mu)$, which contains detailed (though not all) microscopic information, one seeks to describe it in terms of a few macroscopic fields, such as the particle density, momentum density, and energy density, but which we shall denote collectively for now via ϕ . A basic approach would then be to make an ansatz for the distribution function f in terms of these macroscopic fields:

$$f = F[\phi], \quad (11)$$

where F is a functional that maps the macroscopic fields ϕ to a corresponding distribution function f .

One would then ask, after plugging this ansatz into the Boltzmann equation, in what sense it determines the evolution of ϕ . We should well expect that there will be some kind of approximation in the process. After all, solving the Boltzmann equation in general is a hard problem, and we shouldn't expect F to provide an exact solution.

So how do we tell how good an approximate solution is? Consider some candidate for an approximate solution $f_{\text{approx}} = f - \delta f$, where f is a true solution of the Boltzmann equation. Let's consider, in fact, a natural candidate for an approximate solution: a local equilibrium distribution function:

$$f_{\text{approx}} = f_{\text{eq}}(k^\mu; \phi(x^\mu)). \quad (12)$$

The principle here is that the hydrodynamic fields determine at each space-time point a thermodynamic state, and the local equilibrium distribution function is the one that corresponds to this state. We get an estimate for how bad this solution is by plugging it into the Boltzmann equation:

$$0 = k^\mu \partial_\mu f - C[f] = k^\mu \partial_\mu (f_{\text{approx}} + \delta f) - C[f_{\text{approx}} + \delta f] \quad (13)$$

$$\approx k^\mu \partial_\mu (f_{\text{approx}} + \delta f) - C[f_{\text{approx}}] - DC[f_{\text{approx}}][\delta f] \quad (14)$$

$$= k^\mu \partial_\mu (f_{\text{approx}} + \delta f) - DC[f_{\text{approx}}][\delta f]. \quad (15)$$

In the above, we have assumed that δf is small compared to f_{approx} , allowing us to linearize the collision operator around f_{approx} . $DC[f_{\text{approx}}][\delta f]$ represents the collision operator linearized around f_{approx} , applied to δf . By neglecting the δf in the first term, we arrive at

$$DC[f_{\text{approx}}][\delta f] \approx k^\mu \partial_\mu f_{\text{approx}}. \quad (16)$$

Let us now do some dimensional analysis. From the right hand side of the equation, we see that the dimension of DC should be an energy divided by a time scale. The natural choice for the energy is simply the rest-frame energy $u \cdot k$ and the physically expected time scale is the collision time τ_{coll} . On the other hand, the time/space scale of the right hand side is a macroscopic time/length-scale, T (because we are dealing with relativistic fluids, times and lengths are within the same order of magnitude). Since the energy scales on both sides are matched, we conclude that the dimensionless ratio controlling the accuracy of the approximation is

$$\frac{\delta f}{f_{\text{approx}}} \sim \frac{\tau_{\text{coll}}}{T} =: \text{Kn}, \quad (17)$$

where Kn is the Knudsen number.

This then tells us that the correction is of the order of the Knudsen number. If we solve (16) in δf , we can expect an approximate solution for δf , whose error will be of the order of Kn^2 .

Thus we conclude that a good ansatz for F , in cases where a system has a clear separation of scales between the microscopic collision time and the macroscopic evolution time, is given by

$$F[\phi] = f^{(0)}[\phi] + \text{Kn} f^{(1)}[\phi] + \text{Kn}^2 f^{(2)}[\phi] + \dots, \quad (18)$$

where $f^{(0)}$ is the local equilibrium function and the higher-order terms $f^{(n)}[\phi]$ represent successive corrections in powers of the Knudsen number. From a calculation perspective, the higher order terms are determined from the lower order terms, but in principle they depend on the macroscopic fields ϕ and their derivatives. The solution should be off from a true solution by a term of order Kn^{N+1} if we truncate the expansion at order N . This ansatz is the first ingredient in the Chapman-Enskog expansion. The procedure to solve the linearized Boltzman equation, in order to obtain the successive corrections $f^{(n)}[\phi]$ is quite involved. We refer to [5] for a detailed treatment.

1.3 Dynamics of the macroscopic fields

So far we have only constructed approximate solutions to the Boltzmann equation, and have obtained some secondary equations for quantities like δf . There remains the question of what equations govern the macroscopic fields ϕ that appear in the ansatz. For this, we need two things: we need a way to get back the ϕ from an F , and a way to integrate the Boltzmann equation over the k degrees of freedom, in order to 'reduce' them.

In a typical setting, the hydrodynamic fields could be chosen to be the fluid velocity u^μ , the temperature T and the chemical potential μ . It is a bit of a roundabout way that we recover them from F .

Given F , we can first define the energy-momentum tensor as

$$T^{\mu\nu} = \int k^\mu k^\nu F \, dK \quad (19)$$

We then make an informed *choice* that $T^{\mu\nu}$ should encode u^μ as the timelike eigenvector of $T^{\mu\nu}$ corresponding to the eigenvalue ϵ , which is the energy density:

$$u_\mu T^{\mu\nu} = \epsilon u^\nu. \quad (20)$$

This gives us u from F . This choice of u is known as the Landau frame condition.

We next define the particle number current and the particle number density as

$$N^\mu = \int k^\mu F \, dK, \quad n = u_\mu N^\mu. \quad (21)$$

We then *prescribe* that the temperature T and chemical potential μ are such that the particle number density n and energy density ϵ are correctly reproduced by the equilibrium distribution corresponding to T and μ :

$$n = n_{\text{eq}}(T, \mu), \quad \epsilon = \epsilon_{\text{eq}}(T, \mu), \quad (22)$$

where the equilibrium values are obtained in the usual way from the equilibrium distribution function:

$$n_{\text{eq}}(T, \mu) = u_\mu \int k^\mu f_{\text{eq}}(T, \mu) \, dK, \quad \epsilon_{\text{eq}}(T, \mu) = \int (u_\mu k^\mu)^2 f_{\text{eq}}(T, \mu) \, dK. \quad (23)$$

This choice of T and μ is known as the Landau matching condition. It ensures that the moments of the distribution function F corresponding to the particle number density and energy density match those of the equilibrium distribution.

Finally, the hydrodynamic equations are obtained by taking moments of the Boltzmann equation with respect to the conserved quantities, such as particle number and energy-momentum:

$$\partial_\mu N^\mu = 0, \quad \partial_\mu T^{\mu\nu} = 0. \quad (24)$$

For a solution to the Boltzmann equation, these hydrodynamic equations are exactly satisfied: the left hand sides are what one obtains after integrating the Boltzmann equation against the conserved quantities 1 and k^ν , while the right hand sides vanish due to the collision term conserving these quantities.

We take the approximate solution given by a truncation of F , plug it into the definitions of N^μ and $T^{\mu\nu}$, and then into the hydrodynamic equations to obtain approximate equations for the macroscopic fields ϕ . We will see how this works in the subsequent sections.

2 Warm-up: effective collision terms

2.1 The non-relativistic case

While we focus later on the relativistic case, it is useful to begin with the non-relativistic limit, where the Chapman–Enskog mechanism is easier to see. The following subsection gives a schematic BGK derivation of the incompressible Navier–Stokes equation.

The Boltzman equation is

$$\partial_t f + v \cdot \nabla f = \Gamma(f). \quad (25)$$

Here it is understood that $f = f(\vec{x}, \vec{v})$ is the density of particles at a particular position $\vec{x}$ and velocity $\vec{v}$ in phase space. $\Gamma(f)$ is the collision integral, which I won’t even bother to write out, but it is, of course, also a function of $(\vec{x}, \vec{v})$.

The reason I won’t write it out is that I will use a simplified collision term

$$\Gamma(f) \approx -\tau^{-1}(f - f^{eq}), \quad (26)$$

where f^{eq} is some equilibrium distribution we want our solution to converge to. This is known as the BGK model, named after Bhatnagar, Gross and Krook [2], and it is, of course, an approximation, valid under certain conditions. We won’t worry about the conditions, we just see that the r.h.s. has the right dimensions and move on, since modern physics is anyways just

glorified dimensional analysis. This relaxation time τ is supposed to be of the order of the time between two collisions underwent by a single particle, in the hypothesis that after a couple of collisions underwent by a single particle, the system has reached equilibrium.

We do the Chapman-Enskog Ansatz, but only to first order in $\varepsilon = \frac{\lambda}{L}$, the Knudsen number.

$$f = f^{(0)} + \varepsilon f^{(1)}. \quad (27)$$

Here $\lambda \approx v_{thermal}\tau$ is the mean free path, and L is the spatial scale at which significant changes in f occur.

The fact that f changes over distances of order L means that $\nabla f \sim f/L$. To make the Knudsen number appear explicitly in the Boltzman equation, we therefore define the dimensionless operator $\tilde{\nabla} = L\nabla$. Similarly, we define a dimensionless time derivative $\tilde{\partial}_t = T\partial_t$.

The r.h.s. would tempt us to believe that, in time, the relevant changes in f are on scales of order τ . That should be true for microscopic changes, but we can believe that the spatial derivative term induces some kind of slower changes, occurring at times of order T . L and T should be related by the typical velocities corresponding to f . This will justify a model assumption we make further, that the temporal Knudsen number $\frac{\tau}{T}$ equals the spatial one $\frac{\lambda}{L}$.

The Boltzman equation becomes

$$T^{-1}\tilde{\partial}_t f + L^{-1}v \cdot \tilde{\nabla} f = -\tau^{-1}(f - f^{eq}). \quad (28)$$

Or, multiplying by τ and employing the assumption about the equal Knudsen numbers:

$$\tilde{\partial}_t f + \frac{T}{L}v \cdot \tilde{\nabla} f = -\frac{1}{\varepsilon}(f - f^{eq}). \quad (29)$$

We next replace $f = f^{(0)} + \varepsilon f^{(1)}$ and compare order terms, order by order. We get, unsurprisingly, $f^{(0)} = f^{eq}$ and, assuming that f^{eq} is constant in time, $f^{(1)} = -\frac{T}{L}v \cdot \tilde{\nabla} f^{(0)}$, into which if we replace the $\tilde{\nabla}$ we get

$$\varepsilon f^{(1)} = -\tau v \cdot \nabla f^{(0)}. \quad (30)$$

The next step is to look at the momentum conservation equation, which we get out of the Boltzman equation by integrating it against the collisional invariant $\psi = v^i$:

$$\partial_t(nu^i) + \partial_j \int v^j v^i f(\vec{v}) d^3v = 0. \quad (31)$$

After subtracting the continuity equation and doing some standard manipulations, the equation above can be written as

$$\partial_t(nu^i) + u^j \partial_j(nu^i) + \partial_j \left(n \left\langle (v - u)^j (v - u)^i \right\rangle \right) = 0. \quad (32)$$

The last term is

$$\int d^3v \partial_j [f(\vec{v})(v-u)^j(v-u)^i]. \quad (33)$$

It has two contributions, one from $f^{(0)} = f^{eq}$, and another one from $\varepsilon f^{(1)}$. The contribution from $f^{(0)}$ is what we identify as the static pressure (see, for example, [6, Sec. 2.3] for a physical rationale).

We make a modeling choice that n is constant, so that, after dividing by n , this equation becomes:

$$\partial_t u^i + u^j \partial_j u^i + \partial_j (P^{(0)})^{ij} + \frac{1}{n} \partial_j T^{ij} = 0 \quad (34)$$

where $(P^{(0)})^{ij}$ corresponds to the zeroth-order contribution (static pressure) and T^{ij} to the first order. The first three terms correspond to the Euler equation, while, as we will see next, the last term will give us viscosity.

It is time we choose a reasonable expression for $f^{(0)}$. What else but the Maxwell distribution:

$$f^{(0)}(\vec{x}, \vec{v}) = n(\vec{x}) \frac{1}{\mathcal{N}(\beta(\vec{x}))} \exp \left[-\beta(\vec{x}) \frac{m(\vec{v} - \vec{u})^2}{2} \right]. \quad (35)$$

We will, however, make the simplifying assumption that the inverse temperature β and the density n are constant throughout space. Considering temperature and density variations would give us some additional diffusion terms, maybe.

We replace in the expression of $\varepsilon f^{(1)}$

$$\varepsilon f^{(1)} = -\tau \vec{v} \cdot \nabla f^{(0)} = -\tau f^{(0)} \vec{v} \cdot \nabla \log f^{(0)} = -\tau \beta m f^{(0)} v^i (\vec{v} - \vec{u}) \cdot \partial_i \vec{u}. \quad (36)$$

Next we compute

$$T^{ij}(\vec{x}) = \int d^3v \varepsilon f^{(1)}(\vec{x}, \vec{v})(v-u)^i(v-u)^j \quad (37)$$

$$= -\tau \beta m \int d^3v f^{(0)} v^k (v-u)^l \partial_k u^l (v-u)^i (v-u)^j \quad (38)$$

$$= -\tau \beta m n \partial_k u^l \int d^3c f_G(\vec{c}) c^k c^l c^i c^j, \quad (39)$$

where in the last step we used $\vec{c} = \vec{v} - \vec{u}$, $f^{(0)}(\vec{v}) = n f_G(\vec{c})$ and we subtracted $0 = \int d^3c f_G u^k c^l c^i c^j$. Here, $f_G(\vec{c}) = \frac{1}{\mathcal{N}} \exp(-\beta m \frac{\vec{c}^2}{2})$, the gaussian distribution of the peculiar velocity $\vec{c}$.

The integral is

$$\int d^3c f_G(\vec{c}) c^k c^l c^i c^j = (\beta m)^{-2} \left(\delta^{kl} \delta^{ij} + \delta^{ki} \delta^{lj} + \delta^{kj} \delta^{li} \right). \quad (40)$$

After replacing in the expression for T^{ij} and contracting the k, l indices, we are left with

$$T^{ij} = -\tau \frac{k_B T}{m} n \left(\nabla \cdot u \delta^{ij} + \partial_i u^j + \partial_j u^i \right). \quad (41)$$

Next, computing the last term in equation (34), we see

$$\frac{1}{n} \partial_j T^{ij} = -\tau \frac{k_B T}{m} \left(2 \partial_i \nabla \cdot u + \Delta u^i \right), \quad (42)$$

Due to our assumption of incompressibility, $\nabla \cdot u = 0$, so the extra term does not contribute here. More generally, it is the bulk-viscosity contribution: it modifies the effective pressure rather than the shear term, but we chose not to worry about it in this simplified derivation.

Leading to the Navier–Stokes equation

$$\partial_t u^i + u^j \partial_j u^i + \partial_i p - \nu \Delta u^i = 0, \quad (43)$$

where we have used that

$$(P^{(0)})^{ij} = p \delta^{ij}, \quad (44)$$

and we have identified the kinematic viscosity

$$\nu = \tau \frac{k_B T}{m}. \quad (45)$$

Thus, we have related a rough description of microscopic collisions, in terms of τ, T, m to a macroscopic quantity, the kinematic viscosity. What more can you want from life?

2.2 The relativistic case

The Boltzmann-equation with a simplified expression for the collision term in the relativistic case is given by the Anderson–Witting model [1]:

$$k^\mu \partial_\mu f(x, k) = -\frac{u \cdot k}{\tau} (f(x, k) - f_E(u \cdot k)) = -\frac{u \cdot k}{\tau} \delta f(x, k) \quad (46)$$

where $f_E(u \cdot k)$ is the local equilibrium distribution function, which we will make explicit later.

As in the introduction, we can write down an equation for δf by simply ignoring δf on the left-hand side, leading to

$$\delta f(x, k) = -\frac{\tau}{u \cdot k} k^\mu \partial_\mu f_E(u \cdot k). \quad (47)$$

We now continue the idea sketched in the introduction and recover the hydrodynamic fields from f . The local equilibrium, which is zero-order

in the Knudsen number produces the following contribution to the stress tensor:

$$T_0^{\mu\nu} = \int_{\mathbb{R}^4} dK k^\mu k^\nu f_E(u \cdot k) = \int dK \left[u^\mu u^\nu (u \cdot k)^2 f_E(u \cdot k) \right. \quad (48)$$

$$+ u^{(\mu} \Delta_\alpha^{\nu)} k^\alpha (u \cdot k) f_E(u \cdot k) \quad (49)$$

$$+ (\Delta_\alpha^\mu \Delta_\beta^\nu k^\alpha k^\beta) f_E(u \cdot k) \quad (50)$$

$$\left. \right] \quad (51)$$

The mixed spacial-temporal term will vanish due to the isotropy of f_E , and the last integral will equal

$$\int dK (\Delta_\alpha^\mu \Delta_\beta^\nu k^\alpha k^\beta) f_E(u \cdot k) = \Delta^{\mu\nu} \frac{1}{3} \int dK (\Delta_{\alpha\beta} k^\alpha k^\beta) f_E(u \cdot k) \quad (52)$$

$$= \Delta^{\mu\nu} \int dK \mathbf{k}^2 f_E(u \cdot k) =: \Delta^{\mu\nu} P, \quad (53)$$

similarly as in the non-relativistic case.

Thus, to zeroth order in the Knudsen number, the stress tensor is given by the ideal fluid form:

$$T_0^{\mu\nu} = \epsilon u^\mu u^\nu + P \Delta^{\mu\nu}, \quad (54)$$

and the corresponding equation is the relativistic Euler equation:

$$\partial_\mu (\epsilon u^\mu u^\nu + P \Delta^{\mu\nu}) = 0. \quad (55)$$

If particle number is also conserved, the zeroth-order current is

$$N_0^\mu = \int_{\mathbb{R}^4} dK k^\mu f_E = n u^\mu, \quad \partial_\mu N_0^\mu = \partial_\mu (n u^\mu) = 0. \quad (56)$$

The four energy-momentum equations and the particle-number equation evolve n , ϵ , and the three independent components of u^μ .

The remaining quantity P is fixed by the local equilibrium distribution, which depends on the thermodynamic parameters T and μ . Its moments define

$$n = n_{\text{eq}}(T, \mu), \quad \epsilon = \epsilon_{\text{eq}}(T, \mu), \quad P = P_{\text{eq}}(T, \mu). \quad (57)$$

The first two relations are the Landau matching conditions: they determine T and μ from n and ϵ . Substituting them into the last relation yields the equation of state $P = P(\epsilon, n)$, which closes the zeroth-order equations together with $u_\mu u^\mu = 1$.

At first order, the correction (47) modifies the conserved currents through its moments:

$$\delta N^\mu = \int_{\mathbb{R}^4} dK k^\mu \delta f, \quad \delta T^{\mu\nu} = \int_{\mathbb{R}^4} dK k^\mu k^\nu \delta f. \quad (58)$$

The most general decomposition of these corrections relative to u^μ is

$$\delta N^\mu = V^\mu, \quad \delta T^{\mu\nu} = \Pi \Delta^{\mu\nu} + 2u^{(\mu} W^{\nu)} + \pi^{\mu\nu}, \quad (59)$$

where V^μ and W^μ are spatial diffusion and energy-flow currents, Π is the bulk viscous pressure, and the stress tensor $\pi^{\mu\nu}$ is spatial, symmetric, and traceless. In the Landau frame chosen in the introduction, $u_\mu T^{\mu\nu} = \epsilon u^\nu$, so $W^\mu = 0$. In contrast, V^μ need not vanish: it is the particle diffusion current relative to the energy flow.

The Landau matching conditions set the corrections to n and ϵ to zero. In the Landau frame, the first-order conservation laws are

$$\partial_\mu (n u^\mu + V^\mu) = 0, \quad \partial_\mu [\epsilon u^\mu u^\nu + (P + \Pi) \Delta^{\mu\nu} + \pi^{\mu\nu}] = 0. \quad (60)$$

They become the relativistic Navier–Stokes equations once the dissipative currents are expressed to first order in gradients:

$$V^\mu = \kappa \nabla^\mu \left(\frac{\mu}{T} \right), \quad \Pi = -\zeta \theta, \quad \pi^{\mu\nu} = \eta \sigma^{\mu\nu}, \quad (61)$$

where κ , ζ , and η are the diffusion, bulk-viscosity, and shear-viscosity coefficients, respectively.

To illustrate how these transport coefficients arise, we look closer at the shear-stress contribution to $\delta T^{\mu\nu}$, i.e. $\pi^{\mu\nu} = \delta T^{\langle\mu\nu\rangle}$. Inserting (47), the derivative of the local equilibrium distribution gives

$$\pi^{\mu\nu} = \tau \beta \int_{\mathbb{R}^4} dK \frac{1}{u \cdot k} k^{\langle\mu} k^{\nu\rangle} k^\sigma k^\lambda \nabla_\sigma u_\lambda f_E(u \cdot k). \quad (62)$$

We are actually making an extra assumption here that μ and T don't vary, and the derivative only applies to the velocity here. We do this in order to again illustrate the emergence of the shear viscosity. It can be shown that the integral in (47) leads to a result proportional to the shear tensor $\sigma^{\mu\nu}$, yielding the familiar form of the shear-stress tensor in first-order hydrodynamics:

$$\pi^{\mu\nu} = \eta \sigma^{\mu\nu}, \quad \eta = 2\tau\beta A, \quad (63)$$

where η is the shear viscosity, the shear tensor is given by $\sigma^{\mu\nu} = \nabla^{\langle\mu} u^{\nu\rangle}$. The derivation of this result and the coefficient A is given in Appendix A.1.

A Supplementary Calculations

A.1 Shear-stress calculation

Applying the spatial symmetric traceless projection to (47) gives

$$\pi^{\mu\nu} = \tau \beta \int_{\mathbb{R}^4} dK \frac{1}{u \cdot k} k^{\langle\mu} k^{\nu\rangle} k^\sigma k^\lambda \nabla_\sigma u_\lambda f_E(u \cdot k). \quad (64)$$

We first consider the integral

$$\int_{\mathbb{R}^4} dK \frac{1}{u \cdot k} k^\mu k^\nu k^\sigma k^\lambda f_E(u \cdot k) \quad (65)$$

We can evaluate this integral in the local rest frame of the fluid, where $u^\mu = (1, \mathbf{0})$, obtaining

$$\int_{\mathbb{R}^4} dK \frac{1}{u \cdot k} k^\mu k^\nu k^\sigma k^\lambda f_E(u \cdot k) = \Lambda_{\mu'\nu'\sigma'\lambda'}^{\mu\nu\sigma\lambda} \int_{\mathbb{R}^4} dK \frac{1}{k^0} k^{\mu'} k^{\nu'} k^{\sigma'} k^{\lambda'} f_E(k^0), \quad (66)$$

where $\Lambda_{\mu'\nu'\sigma'\lambda'}^{\mu\nu\sigma\lambda} = \Lambda_{\mu'}^\mu \Lambda_{\nu'}^\nu \Lambda_{\sigma'}^\sigma \Lambda_{\lambda'}^\lambda$ transforms vectors from the local rest frame to whatever frame we were expressing u in before.

We can generally decompose $k^{\mu'} = (u \cdot k)u^{\mu'} + \Delta_{\nu'}^{\mu'} k^{\nu'}$. In the case of a rest frame, $u^{\mu'} = (1, \mathbf{0})$, and

$$\Delta_{\nu'}^{\mu'} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

so the decomposition becomes

$$k^{\mu'} = (k^0)u^{\mu'} + \Delta_i^{\mu'} k^i. \quad (67)$$

Replacing the decomposition of $k^{\mu'}$ in the integral, we obtain a purely temporal term, a purely spatial term, and mixed terms. Let's write them out:

$$k^{\mu'} k^{\nu'} k^{\sigma'} k^{\lambda'} = \Delta_i^{\mu'} \Delta_j^{\nu'} \Delta_k^{\sigma'} \Delta_l^{\lambda'} k^i k^j k^k k^l \quad (68)$$

$$+ (k^0)^2 (u^{\mu'} u^{\nu'} \Delta_{\rho'}^{\sigma'} k^{\rho'} \Delta_{\tau'}^{\lambda'} k^{\tau'} + \text{permutations}) \quad (69)$$

$$+ (k^0)^4 u^{\mu'} u^{\nu'} u^{\sigma'} u^{\lambda'} \quad (70)$$

$$+ (k^0) (u^{\mu'} \Delta_i^{\nu'} \Delta_j^{\sigma'} \Delta_k^{\lambda'} k^i k^j k^k + \text{permutations}) \quad (71)$$

$$+ (k^0)^3 (u^{\mu'} u^{\nu'} u^{\sigma'} \Delta_{\rho'}^{\lambda'} k^{\rho'} + \text{permutations}). \quad (72)$$

We have left the terms with an odd number of spacial indices at the end on purpose. These terms will vanish upon integration. The integral therefore becomes:

$$\begin{aligned} & \int_{\mathbb{R}^4} dK \frac{1}{k^0} k^{\mu'} k^{\nu'} k^{\sigma'} k^{\lambda'} f_E(k^0) = \\ & \Delta_i^{\mu'} \Delta_j^{\nu'} \Delta_k^{\sigma'} \Delta_l^{\lambda'} \int_{\mathbb{R}^4} dK \frac{1}{k^0} k^i k^j k^k k^l f_E(k^0) \\ & + \left(\Delta_i^{\mu'} \Delta_j^{\nu'} u^{\sigma'} u^{\lambda'} + 5 \text{ perms.} \right) \int_{\mathbb{R}^4} dK \frac{1}{k^0} (k^0)^2 k^i k^j f_E(k^0) \\ & = A \Delta_i^{\mu'} \Delta_j^{\nu'} \Delta_k^{\sigma'} \Delta_l^{\lambda'} (\Delta^{ij} \Delta^{kl} + \Delta^{ik} \Delta^{jl} + \Delta^{il} \Delta^{jk}) + B \left(\Delta_i^{\mu'} \Delta_j^{\nu'} u^{\sigma'} u^{\lambda'} + 5 \text{ perms.} \right) \Delta^{ij} \\ & = A (\Delta^{\mu'\nu'} \Delta^{\sigma'\lambda'} + \Delta^{\mu'\sigma'} \Delta^{\nu'\lambda'} + \Delta^{\mu'\lambda'} \Delta^{\nu'\sigma'}) + B (\Delta^{\mu'\nu'} u^{\sigma'} u^{\lambda'} + 5 \text{ perms.}) \end{aligned}$$

To determine A , contract the purely spatial fourth moment with $\delta_{ij}\delta_{kl}$. The three tensor structures on the right-hand side each contribute, giving a factor of $3^2 + 3 + 3 = 15$. Hence,

$$A = \frac{1}{15} \int_{\mathbb{R}^4} dK \frac{\mathbf{k}^4}{k^0} f_E(k^0). \quad (73)$$

After multiplying by $\Lambda_{\mu'\nu'\sigma'\lambda'}^{\mu\nu\sigma\lambda}$, the integral simplifies to

$$\Lambda_{\mu'\nu'\sigma'\lambda'}^{\mu\nu\sigma\lambda} \int_{\mathbb{R}^4} dK \frac{1}{k^0} k^{\mu'} k^{\nu'} k^{\sigma'} k^{\lambda'} f_E(k^0) = A (\Delta^{\mu\nu} \Delta^{\sigma\lambda} + \Delta^{\mu\sigma} \Delta^{\nu\lambda} + \Delta^{\mu\lambda} \Delta^{\nu\sigma}) + B (\Delta^{\mu\nu} u^\sigma u^\lambda + 5 \text{ perms.}).$$

Furthermore, once we take the spatial symmetric traceless projection, the term proportional to B will vanish, leaving only the term proportional to A . This is because $\Delta^{\langle\mu\nu\rangle} = 0$, and the other permutations contain either u^μ or u^ν , which also vanish under the projection.

We therefore obtain

$$\pi^{\mu\nu} = \tau\beta A \Delta^{\sigma(\mu} \Delta^{\nu)\lambda} (\nabla_\sigma u_\lambda + \nabla_\lambda u_\sigma). \quad (74)$$

Using the standard definition of the shear tensor,

$$\theta := \Delta^{\alpha\beta} \nabla_\alpha u_\beta, \quad \theta = \nabla_\rho u^\rho \quad \text{when } u_\rho u^\rho = 1, \quad (75)$$

where the second equality follows from $u^\alpha u_\beta \nabla_\alpha u^\beta = 0$. In three spatial dimensions,

$$\sigma^{\mu\nu} := \Delta^{\mu\alpha} \Delta^{\nu\beta} \left[\frac{1}{2} (\nabla_\alpha u_\beta + \nabla_\beta u_\alpha) - \frac{1}{3} \Delta_{\alpha\beta} \theta \right], \quad (76)$$

where the angle brackets denote the spatial symmetric traceless projection. Therefore,

$$\Delta^{\sigma(\mu} \Delta^{\nu)\lambda} (\nabla_\sigma u_\lambda + \nabla_\lambda u_\sigma) = 2\sigma^{\mu\nu}, \quad (77)$$

and the stress tensor is proportional to the shear tensor:

$$\pi^{\mu\nu} = 2\tau\beta A \sigma^{\mu\nu}. \quad (78)$$

Returning to the computation of A , we have

$$A = \frac{1}{15} \int_{\mathbb{R}^3} \frac{d^3 k}{(2\pi)^3} \frac{\mathbf{k}^4}{k^0} f_E(k^0) \quad (79)$$

$$= \frac{1}{15} \int_{\mathbb{R}^3} \frac{d^3 k}{(2\pi)^3} \frac{\mathbf{k}^4}{\sqrt{\mathbf{k}^2 + m^2}} \exp\left(\alpha - \frac{\sqrt{\mathbf{k}^2 + m^2}}{T}\right). \quad (80)$$

Using spherical coordinates,

$$d^3k = 4\pi k^2 dk,$$

we obtain

$$A = \frac{e^\alpha}{15} \frac{4\pi}{(2\pi)^3} \int_0^\infty dk, \frac{k^6}{\sqrt{k^2 + m^2}} \exp\left(-\frac{\sqrt{k^2 + m^2}}{T}\right) \quad (81)$$

$$= \frac{e^\alpha}{30\pi^2} \int_0^\infty dk, \frac{k^6}{\sqrt{k^2 + m^2}} \exp\left(-\frac{\sqrt{k^2 + m^2}}{T}\right). \quad (82)$$

This integral can be evaluated in terms of modified Bessel functions of the second kind, but the details are not particularly illuminating for the present discussion. We just note that the result will be of the form:

$$A = e^{\mu/T} m^6 K\left(\frac{m}{T}\right), \quad (83)$$

for some non-dimensional function $K(m/T)$.

References

- [1] John L. Anderson and Henry R. Witting, *A relativistic relaxation-time model for the Boltzmann equation*, Physica **74** (1974), no. 3, 466–488.
- [2] Prabhu L. Bhatnagar, Eugene P. Gross, and Max Krook, *A model for collision processes in gases*, Physical Review **94** (1954), no. 3, 511–525.
- [3] Carlo Cercignani, *The boltzmann equation and its applications*, Springer, 1988.
- [4] Sydney Chapman and T. G. Cowling, *The mathematical theory of non-uniform gases*, 3 ed., Cambridge University Press, 1970.
- [5] Gabriel S. Denicol, *Kinetic foundations of relativistic dissipative fluid dynamics*, Journal of Physics G: Nuclear and Particle Physics **41** (2014), no. 12, 124004.
- [6] Stewart Harris, *An introduction to the theory of the Boltzmann equation*, Dover Publications, 2004, Originally published by Holt, Rinehart and Winston, 1971.